

SMART RECRUITMENT MATCHING PLATFORM

How to Run the Project

Setup, Demo Credentials, and Recommended User Journey

Quick Start

1. Open the project folder in Visual Studio 2022.
2. Restore NuGet packages and create/update the LocalDB database.
3. Run the project with the HTTPS profile.
4. Open <https://localhost:7286/> for the web interface.
5. Register a Job Seeker and Employer account, then follow the demo flow below.

1. Project Overview

Smart Recruitment Matching Platform is an ASP.NET Core Web API application with a built-in HTML/CSS/JavaScript frontend. The backend and frontend are served by the same ASP.NET Core project. The final project targets .NET 8 and uses Entity Framework Core with SQL Server LocalDB.

2. Prerequisites

- Windows 10/11
- Visual Studio 2022 with the ASP.NET and web development workload
- .NET 8 SDK
- SQL Server Express LocalDB (normally installed with Visual Studio)
- Internet connection only for initial NuGet package restore, if packages are not already cached

3. Open the Project

Extract the submitted ZIP file, then open the project using either of these options:

- Open SmartRecruitment-Project.slnx in Visual Studio 2022.
- Or open a terminal in the folder that contains SmartRecruitment-Project.csproj.

4. Restore Dependencies

Visual Studio normally restores NuGet packages automatically. If needed, run:

```
dotnet restore
```

5. Create / Update the Database

The project is configured to use SQL Server LocalDB with database name SmartRecruitmentDB. The final ZIP contains Entity Framework Core migrations, so apply them before the first run.

Recommended Visual Studio method:

- Open Tools > NuGet Package Manager > Package Manager Console.
- Make sure the SmartRecruitment project is selected as the Default project.
- Run:

Update-Database

Command-line alternative:

dotnet ef database update

If the dotnet ef command is not available, use Package Manager Console as described above.

6. Run the Application

In Visual Studio:

- Select the HTTPS launch profile.
- Press F5 (Debug) or Ctrl+F5 (Run without debugging).

Command-line alternative:

dotnet run --launch-profile https

Default local addresses from the final project configuration:

Purpose	URL
Main web application	https://localhost:7286/
Swagger / API testing	https://localhost:7286/swagger
HTTP fallback	http://localhost:5022/

If the browser warns about the local HTTPS development certificate, trust the .NET development certificate in Visual Studio, or run the following once:

dotnet dev-certs https --trust

7. Login Credentials and First-Time Demo Accounts

Important: the submitted source code does not contain hard-coded or seeded user passwords. Job Seeker and Employer accounts are intentionally created through the registration pages. For an easy assessor demo, create the following accounts once after the database is created.

Role	Demo Email	Demo Password	How to Create
Job Seeker	jobseeker.demo@smart.local	Demo@123	Register from the Job Seeker registration page
Employer	employer.demo@smart.local	Demo@123	Register from the Employer registration page
Administrator	admin.demo@smart.local	Demo@123	Create once using the admin setup steps below

The password Demo@123 satisfies the project's password rule: at least 8 characters with uppercase, lowercase, number, and special character.

8. One-Time Administrator Account Setup

The final project provides Administrator login and admin pages, but it does not provide a public Administrator registration endpoint or a seeded administrator account. For local evaluation, create the demo administrator once as follows:

- Open the Job Seeker registration page and register admin.demo@smart.local with password Demo@123.
- Open SQL Server Object Explorer / SSMS and connect to (localdb)\MSSQLLocalDB.
- Open the SmartRecruitmentDB database and run the SQL statement below.

```
UPDATE Users SET Role = 3 WHERE Email = 'admin.demo@smart.local';
```

Role value 3 is Administrator in the project's UserRole enum. Log out after the update and sign in again with admin.demo@smart.local / Demo@123.

9. Recommended Evaluation / User Journey

For the easiest end-to-end demonstration, use the following order. This creates the data needed by later steps and lets the assessor see the full recruitment workflow without manually preparing records.

A. Employer Flow - Create the Job First

1. Go to the home page and choose 'Post Jobs as Employer' (or open the Employer registration page).
2. Register employer.demo@smart.local using Demo@123.
3. After login, open Employer Profile and complete the company information.
4. Open Vacancies and create a vacancy.
5. Enter title, description, location, minimum experience, required education, and required skills.
6. Keep the vacancy Open.

7. The employer can later return to Vacancies to edit or close the vacancy.

B. Job Seeker Flow - Profile, Matching, and Application

1. Log out from the Employer account.
2. Register jobseeker.demo@smart.local using Demo@123.
3. Open Job Seeker Profile.
4. Enter full name, location, years of experience, education level, summary, and skills.
5. Upload a CV from the profile page.
6. Open Jobs to browse the Employer's open vacancy.
7. Open Job Details to view the calculated match score, required skills, and missing skills.
8. Click Apply. The system stores the application and prevents applying to the same vacancy twice.
9. Open My Applications to see the application status and match score.

C. Employer Flow - Review the Applicant

1. Log out and sign in again as employer.demo@smart.local / Demo@123.
2. Open Vacancies and select the created vacancy.
3. Open Applicants. Applicants are displayed in match-score ranking order.
4. Review the candidate information and, where available, download/view the applicant CV.
5. Update the application status (for example Under Review, Shortlisted, Accepted, or Rejected).
6. Optionally send a Contact Request to the candidate.

D. Job Seeker Flow - Notifications and Contact Request

1. Log out and sign in as jobseeker.demo@smart.local / Demo@123.
2. Open My Applications to confirm the new application status.
3. Open Notifications to view the application-status notification.
4. Open Contact Requests to view the Employer request.
5. Accept or decline the request.

E. Employer Flow - Confirm Contact Response

1. Sign in again as employer.demo@smart.local / Demo@123.
2. Open Notifications / Contact Requests.
3. Confirm that the Job Seeker's response is visible.

F. Administrator Flow

1. Sign in as admin.demo@smart.local / Demo@123 after completing the one-time Administrator setup.
2. Open the Admin Dashboard to view total users, vacancies, and applications.
3. Open Users to review accounts.
4. Use Activate / Deactivate to demonstrate account management.

10. Optional API Testing with Swagger

Swagger is enabled in the Development environment. Open <https://localhost:7286/swagger>. Registration and login endpoints are public. For protected endpoints:

- Call POST /api/auth/login using one of the demo accounts.
- Copy the JWT token returned in the response.
- Click Authorize in Swagger.
- Enter the token in the Bearer authentication dialog.
- Call endpoints that match the logged-in user's role.

11. Common Startup Problems

Problem	Fix
Database/table not found	Run Update-Database or dotnet ef database update.
HTTPS certificate warning	Run dotnet dev-certs https --trust, then restart the browser/project.
Port 7286 is already in use	Stop the other process or run from Visual Studio using an available profile; if the HTTPS port changes, update wwwroot/js/config.js accordingly.
401 Unauthorized	Log in again and ensure the JWT has not expired. Tokens are configured for 60 minutes.
403 Forbidden	Use an account with the correct role for that page/API.
Frontend cannot call API	Run the frontend through the ASP.NET Core project URL, not by double-clicking HTML files from disk.

12. Final Quick Checklist for the Assessor

- Database migration completed.
- Application opens at <https://localhost:7286/>.
- Employer can create an open vacancy.
- Job Seeker can complete profile, add skills, upload CV, view match details, and apply.
- Employer can see ranked applicants and update application status.
- Notifications and contact-request workflow can be demonstrated.
- Administrator dashboard and user management can be demonstrated after one-time admin setup.

Prepared from the submitted final project ZIP and its actual configuration.